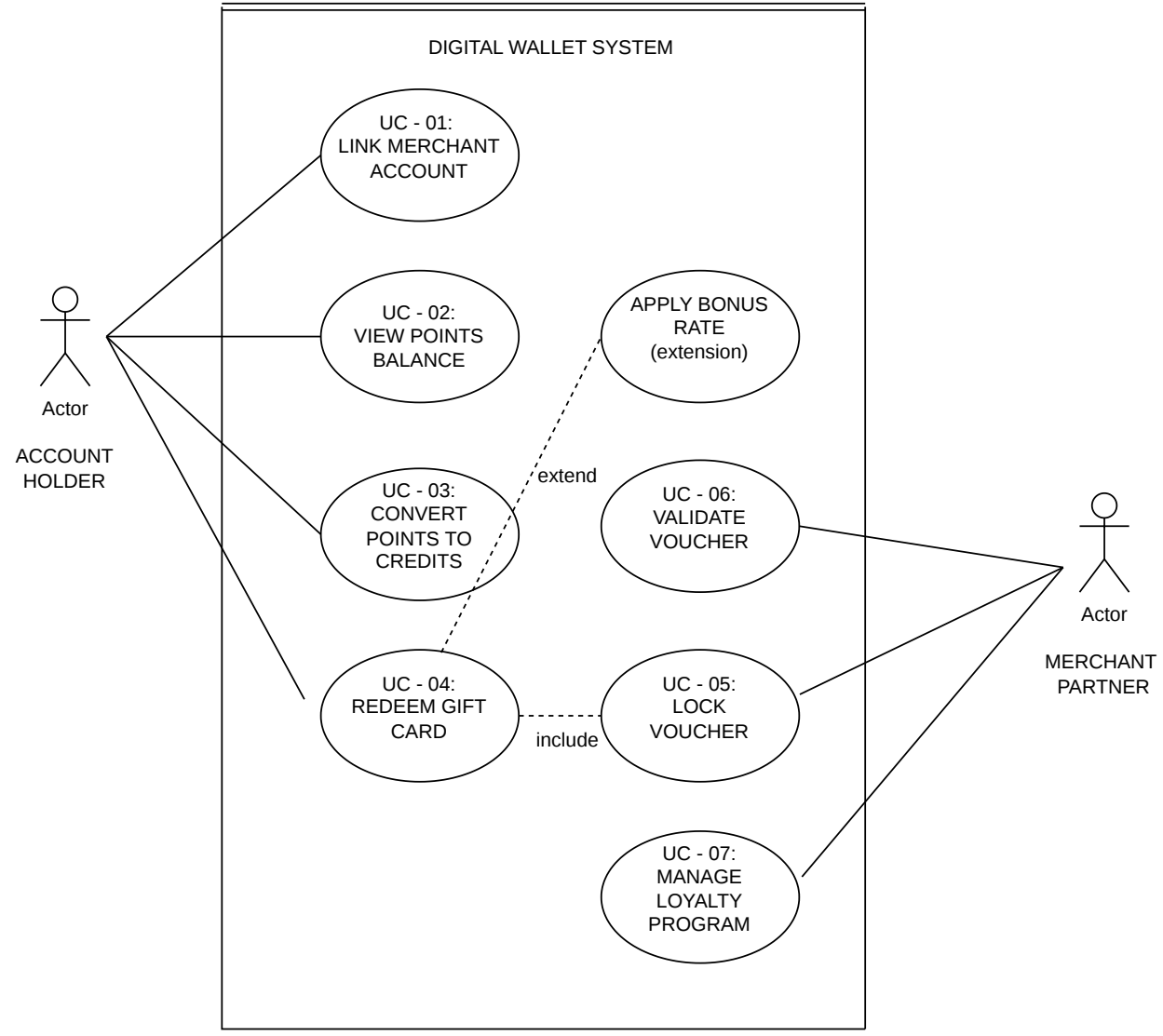
Requirements Table — Digital Gift Card & Loyalty Point Exchange

Problem Statement 34 — Retail, E-Commerce & Finance

Req ID	Type	Description	Priority	Acceptance Criteria	Rationale
FR-001	Functional	The system shall validate loyalty points submitted for conversion against the originating merchant's active ledger, convert valid points into universal exchange credits, and generate a single-use digital voucher code with a cryptographic checksum.	High	Pass: a voucher is generated only when checksum validation succeeds and the source points are marked as spent. Fail: a voucher is generated for expired, already-redeemed, or unverified points.	Core conversion and anti-fraud functionality; the platform's primary value proposition.
FR-002	Functional	The system shall allow an Account Holder to link one or more Merchant Partner loyalty accounts and view their aggregated point balance in a single wallet view.	High	When a valid merchant account is linked, its point balance appears in the aggregated view within 5 seconds; invalid credentials produce an error and no balance is added.	Aggregation across brands is the central value proposition of the wallet.
FR-003	Functional	The system shall apply the current dynamic conversion rate when converting loyalty points to universal exchange credits, and shall display that rate to the Account Holder for confirmation before conversion.	Medium	The rate shown at confirmation matches the rate applied to the transaction to within 0.01%; conversion cannot proceed without explicit Account Holder confirmation.	Rate transparency prevents disputes and builds trust in a fluctuating exchange model.
FR-004	Functional	The system shall lock a gift card voucher immediately upon generation so that it cannot be redeemed a second time.	High	Any attempt to redeem an already-locked voucher is rejected 100% of the time and the attempt is logged with a timestamp and merchant ID.	Directly implements the anti-fraud voucher locking requirement central to the platform.
FR-005	Functional	The system shall allow a Merchant Partner to validate a presented voucher code against the platform ledger before honoring redemption.	High	A merchant-side validation call returns a Valid/Invalid status within 250 ms; invalid, expired, or already-used vouchers return a Fail status.	Enables real-world redemption at the merchant point of sale and closes the loop on fraud prevention.
NFR-001	Nonfunctional	Point conversion and gift card redemption operations shall complete within 250 ms with zero ledger balance drift, even under peak load.	High	Benchmark tests under simulated peak load confirm sub-250 ms latency for 95% of transactions and ledger totals reconcile with zero drift.	Latency and ledger integrity are essential to user trust and financial accuracy.
NFR-002	Nonfunctional	The system shall encrypt all loyalty point, voucher, and transaction data both in transit and at rest using industry-standard encryption (TLS 1.2+ and AES-256).	High	A security audit confirms no loyalty, voucher, or transaction data is transmitted or stored without encryption; any unencrypted channel or store is flagged as a failure.	Protects Account Holder and Merchant Partner financial data and supports the platform's anti-fraud commitments.


DIGITAL WALLET SYSTEM
UC - 01: LINK MERCHANT ACCOUNT
UC - 02: VIEW POINTS BALANCE
UC - 03: CONVERT POINTS TO CREDITS
UC - 04: REDEEM GIFT CARD
APPLY BONUS RATE (extension)
extend
UC - 06: VALIDATE VOUCHER
UC - 05: LOCK VOUCHER
include
UC - 07: MANAGE LOYALTY PROGRAM
Actor
ACCOUNT HOLDER
Actor
MERCHANT PARTNER

Use-Case Flow: Redeem Digital Gift Card

Problem Statement #34 — Digital Gift Card & Loyalty Point Exchange

Use Case

UC-04: Redeem Digital Gift Card (includes UC-05: Lock Voucher)

Actors

Primary Actor: Account Holder — initiates the redemption and receives the gift card voucher.

Secondary Actor: Merchant Partner — validates and honors the voucher at the point of sale.

Preconditions

1. The Account Holder is logged into the wallet platform with at least one linked Merchant Partner loyalty account.
2. The Account Holder has a sufficient converted exchange-credit balance to cover the gift card value requested.

Postconditions

Success: A single-use, checksum-verified voucher is generated and locked, the Account Holder's exchange-credit balance is debited, and the transaction is recorded in the ledger with zero balance drift.

Failure: No voucher is generated, no balance is debited, and the failed attempt is logged.

Main Success Scenario

1. Account Holder selects "Redeem Gift Card" and chooses a Merchant Partner and denomination.
2. System checks the Account Holder's converted exchange-credit balance against the requested amount.
3. System validates the request and reserves the corresponding credits.
4. System generates a single-use digital voucher code with a cryptographic checksum.
5. System immediately locks the voucher so it cannot be redeemed a second time — «include» UC-05: Lock Voucher.
6. System debits the Account Holder's exchange-credit balance and records the transaction in the ledger.
7. System displays the voucher code and sends a confirmation notification to the Account Holder.
8. At the point of sale, the Merchant Partner submits the voucher code for validation.
9. System confirms the voucher is valid, unused, and unexpired, and returns a "Valid" status within 250 ms.
10. Merchant Partner honors the redemption; system marks the voucher as redeemed.
11. Use case ends successfully.

Alternate Flow

2a. Insufficient Balance

2a1. System displays an "Insufficient Balance" message and shows the current exchange-credit balance.

2a2. Account Holder is prompted to convert additional loyalty points (UC-03) or reduce the requested denomination.

2a3. If the Account Holder does not take either action, the redemption request is cancelled and no credits are reserved.

Alternate Flow (Merchant-Side)

9a. Voucher Fails Validation

9a1. System returns an "Invalid" status if the voucher is expired, already redeemed, or fails checksum verification.

9a2. Merchant Partner declines the redemption and informs the Account Holder.

9a3. System logs the failed validation attempt, including merchant ID and timestamp, for fraud review.